

DSA / Algorithms (Abdul Bari) - Day 3

45-day plan mapped to the Algorithms playlist

Time Complexity

Deriving time complexity of loops, nested loops, while and if.

Watch these Abdul Bari videos

- > 1.5.1 Time Complexity #1
- > 1.5.2 Time Complexity Example #2
- > 1.5.3 Time Complexity of While and if #3

Key concepts

- Single loop $\rightarrow O(n)$; nested loops $\rightarrow O(n^2)$.
- While loops depend on how the variable changes (often $O(\log n)$).
- Keep only the dominant term.

Key idea

```
for i: for j  $\rightarrow O(n^2)$ ; while( $i*=2$ )  $\rightarrow O(\log n)$ .
```

Tasks (1.5h)

- [] Watch the 3 videos
- [] Find complexity of 3 loop patterns
- [] Note the dominant term rule

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0